Seyed Soheil Johari

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Professional Summary

Full-time industry experience as a **Data Center Networking Researcher** at **Huawei Technologies**, with expertise in **large-scale AI infrastructure**, **GPU-cluster networking**, **RDMA/RoCEv2**, **congestion control**, **load balancing**, and performance bottleneck analysis for **distributed AI training** and **inference** workloads. My work focuses on designing and evaluating **high-performance networking** mechanisms for **multi-node GPU clusters**, including transport-level optimization, workload-driven performance analysis, and infrastructure designs that improve scalability, bandwidth utilization, and low-latency communication. I also have a Ph.D. research background in Computer Science at the **University of Waterloo**, with experience in AI-driven fault management for **cloud-native networked systems**.

Education

Ph.D. in Computer Science, ***University of Waterloo*** — **2021 — 2026**

B.Sc. in Electrical Engineering, ***Sharif University of Technology*** — **2015 — 2020**

Technical and Industrial Experience

DCN Researcher: AI Infrastructure Networking
Huawei Technologies — **May 2025 – April 2026**

- Optimization of **large-scale AI training GPU cluster** networking for **Large Language Model (LLM)** training and inference workloads, focusing on load balancing, congestion control and other **transport-layer** mechanisms in **RDMA-based** data centers.
- Designed a new **load balancing** mechanism for ACK coalescing scenarios in AI training networks, improving **bandwidth utilization by up to 40%** under **link failure** scenarios compared to **UEC-standardized** schemes. **patent application under review.**
- Designed an **in-switch load balancing architecture** at the **destination core switch**, separating source-DC and destination-DC decisions and improving **bandwidth utilization by up to 24%** under **link failure** scenarios. **patent application under review.**
- Designed a novel **credit-based host-side congestion control** mechanism for AI training clusters, improving **flow completion time by up to 15%** under bursty **distributed training** workloads.
- Designed realistic **AI training and inference workloads** for Huawei's AI training environment to evaluate transport-layer mechanisms (**all-reduce**, **bursty gradients**, heterogeneous flows, pipeline **parallelism**, and **latency-sensitive inference**).
- Contributed to the design and refinement of *HT-CC*, a lightweight **congestion control** algorithm optimized for ease of deployment in **Huawei AI clusters**, focusing on simplicity, stability, and practical implementability.

Research Assistant: AI-driven 5G Network Slice Operations and Management
University of Waterloo — **January 2021 – April 2025**

- Project page: https://rboutaba-cs.github.io/wat5gslicing/
- Collaborated with Rogers on **AI-driven 5G network automation**, focusing on telemetry-based **fault detection, analysis, and resolution** in **cloud-native**, **NFV**-based network slices.
- Proposed a **causal RCA framework** for 5G and cloud-native systems, learning a **latent-aware structural prior** from normal telemetry and improving RCA accuracy by up to **38%**.
- Developed an **Active Learning** framework for **Transformer**-based fault diagnosis, selecting informative telemetry samples while maintaining accuracy with **50% fewer labels**.
- Proposed a robust **anomaly detection and localization** framework for NFV systems using **masked autoencoders**, **Noisy-Student training**, and **XAI**, improving detection by up to 24% and localization by 22%.

- Developed **Deep Reinforcement Learning**-based re-optimization for network slice embedding in **Elastic Optical Networks**, reducing spectrum fragmentation while minimizing reconfiguration overhead.

MITACS Research Intern: ML-driven management of 5G Networks
Rogers Communications **May 2021 – December 2024**

- Contributed to **industry-facing** implementation and validation of AI-driven **fault and performance management** solutions for softwarized, virtualized 5G network slices with Rogers engineers.
- Processed and analyzed real and emulated **5G core network telemetry** (KPIs, logs, alarms, resource metrics) to support model development, evaluation, and benchmarking under **realistic operational** scenarios.
- Implemented end-to-end **data pipelines** and experimental frameworks for anomaly detection, fault diagnosis, and root cause analysis in cloud-native, NFV-based environments.

Skills

Networking	High-Performance Data Center Networking for AI/ML Clusters, RDMA/RoCEv2, InfiniBand concepts, NVLink/NVSwitch concepts, Congestion Control, Load Balancing, Transport Protocol Design, Collective Communication, AllReduce Workload Modeling, Ultra Ethernet (UEC concepts), NS3, Linux Networking, Clos / Fat-tree topologies, Programmable Switches, SDN, NFV, Network Softwarization,
Machine Learning	Distributed Training Optimization for LLM Workloads, Distributed Inference and Serving, vLLM, NCCL performance tuning, Communication-Efficient ML (AllReduce, Collective Optimization), AI Training and Inference Workload Modeling, Generative AI, Multi-Agent and Multimodal AI Systems, Applied Machine Learning
Cloud and Infrastructure	Large-Scale AI Infrastructure, Multi-node GPU Cluster Concepts, Kubernetes at Scale, Docker, Containers, Helm, Infrastructure as Code (Terraform), Multi-cloud Infrastructure (AWS / Azure), Scalable ML Infrastructure Design, Linux Fundamentals, Node-level Performance Analysis, System Observability (Prometheus, Grafana), Performance Debugging and Bottleneck Analysis, Automation and Deployment Pipelines
Programming	C, C++, CUDA, Python, PyTorch, PyTorch Geometric, TensorFlow, Pandas, Java, Git, MATLAB, VHDL, Verilog

Publications

- **S. S. Johari**, R. Boutaba. "Root Cause Analysis in Networked Systems via Latent Cause Inference and Prior Knowledge." *Submitted to ACM SIGMETRICS*. Under Review.
- **S. S. Johari**, M. Tornatore, R. Boutaba, A. Saleh. "Few-Shot Domain Adaptation for Effective Data Drift Mitigation in Network Management." *IEEE International Conference on Distributed Computing Systems (ICDCS) 2025.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Anomaly Detection and Localization in NFV Systems by Utilizing Masked-Autoencoder and XAI." *IEEE Transactions on Mobile Computing (TMC) 2025.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Active Learning for Transformer-Based Fault Diagnosis in 5G and Beyond Mobile Networks." *IEEE Transactions on Network and Service Management (TNSM) 2025.* [pdf]
- **S. S. Johari**, S. Taeb, N. Shahriar, S. R. Chowdhury, M. Tornatore, R. Boutaba. "DRL-Assisted Reoptimization of Network Slice Embedding on EON-enabled Transport Networks." *IEEE Transactions on Network and Service Management (TNSM) 2023.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Anomaly Detection and Localization in NFV Systems: An Unsupervised Learning Approach." *IEEE/IFIP Network Operations and Management Symposium (NOMS) 2022.* **Recipient of the Best Student Paper Award.** [pdf]